

CHAPTER 11

Photosynthesis in Higher Plants

Experiments of Photosynthesis

1. Anoxygenic photosynthesis is characteristic of: (2014)
 a. *Ulva* b. *Rhodospirillum*
 c. *Spirogyra* d. *Chlamydomonas*

Location & Pigments of Photosynthesis

2. Match List-I with List-II. (2023-Manipur)

List-I		List-II	
(A)	Chlorophyll <i>a</i>	(I)	Yellow to yellow orange
(B)	Chlorophyll <i>b</i>	(II)	Yellow green
(C)	Xanthophyll	(III)	Blue green
(D)	Carotenoid	(IV)	Yellow

Choose the **correct** answer from the options given below:

- a. A-III, B-II, C-IV, D-I b. A-III, B-I, C-IV, D-II
 c. A-II, B-III, C-I, D-IV d. A-IV, B-III, C-II, D-I
3. Which of the following statements is incorrect? (2021)
 a. Stroma lamellae have PS I only and lack NADP reductase.
 b. Grana lamellae have both PS I and PS II.
 c. Cyclic photophosphorylation involves both PS I and PS II.
 d. Both ATP and NADPH + H⁺ are synthesized during non-cyclic photophosphorylation.
4. Phytochrome is a: (2016 - II)
 a. Lipoprotein b. Chromoprotein
 c. Flavoprotein d. Glycoprotein
5. In a chloroplast the highest number of protons are found in: (2016 - I)
 a. Stroma b. Lumen of thylakoid
 c. Inter membranal space d. Antennae complex
6. Water soluble pigments found in plant cell vacuoles are: (2016 - I)
 a. Xanthophylls b. Chlorophylls
 c. Carotenoids d. Anthocyanins
7. Emerson's enhancement effect and Red drop have been instrumental in the discovery of: (2016 - I)
 a. Photophosphorylation and non-cyclic electron transport
 b. Two photosystem operating simultaneously
 c. Photophosphorylation and cyclic electron transport
 d. Oxidative phosphorylation

8. In photosynthesis, the light-independent reactions take place at: (2015 Re)
 a. Photosystem-I b. Photosystem-II
 c. Stromal matrix d. Thylakoid lumen

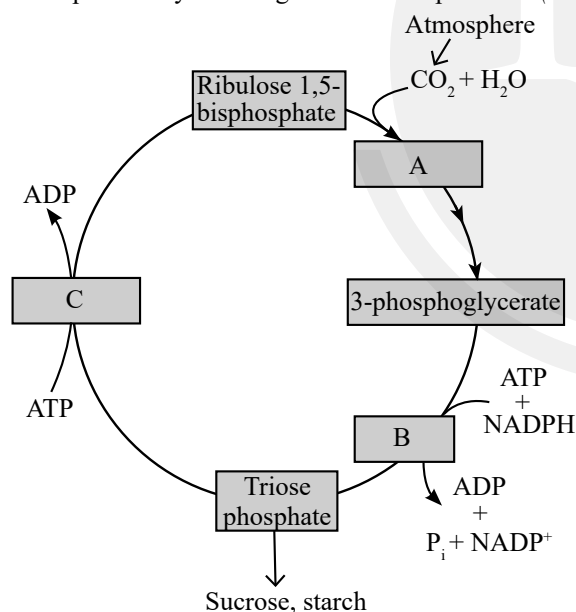
Light Reaction & Electron Transport

9. Which of the following are required for the light reaction of Photosynthesis? (2024 Re)
 A. CO₂ B. O₂ C. H₂O
 D. Chlorophyll E. Light
 Choose the **correct** answer from the options given below:
 a. A, C, D and E only b. C, D and E only
 c. A and B only d. A, C and E only
10. The reaction centre in PS II has an absorption maxima at (2023)
 a. 780 nm b. 680 nm c. 700 nm d. 660 nm
11. Which of the following combinations is required for chemiosmosis? (2023)
 a. proton pump, electron gradient, NADP synthase
 b. membrane, proton pump, proton gradient, ATP synthase
 c. membrane, proton pump, proton gradient, NADP synthase
 d. proton pump, electron gradient, ATP synthase
12. Which out of the following statements is incorrect? (2023-Manipur)
 a. Grana lamellae have both PS I and PS II
 b. Cyclic photophosphorylation involves both PS I and PS II
 c. Both ATP and NADPH + H⁺ are synthesised during non-cyclic photophosphorylation
 d. Stroma lamellae lack PS II and NADP reductase
13. Identify the correct statements regarding chemiosmotic hypothesis (2022 Re)
 A. Splitting of the water molecule takes place on the inner side of the membrane.
 B. Protons accumulate within the lumen of the thylakoids.
 C. Primary acceptor of electron transfers the electrons to an electron carrier.
 D. NADP reductase enzyme is located on the stroma side of the membrane.
 E. Protons increase in number in stroma.
 Choose the correct answer from the options given below:
 a. (B), (C) and (E) b. (A), (B) and (E)
 c. (A), (B) and (D) d. (B), (C) and (D)

14. Which one of the following is **not true** regarding the release of energy during ATP synthesis through chemiosmosis? It involves: (2022)
- Reduction of NADP to NADPH₂ on the stroma side of the membrane
 - Breakdown of proton gradient
 - Breakdown of electron gradient
 - Movement of protons across the membrane to the stroma
15. In light reaction, plastoquinone facilitates the transfer of electrons from: (2020)
- Cytb₆f complex to PS-I
 - PS-I to NADP⁺
 - PS-I to ATP synthase
 - PS-II to Cytb₆f complex
16. During non-cyclic photophosphorylation, when electrons are lost from the reaction centre at PS II, what is the source which replaces these electrons? (2020-Covid)
- Water
 - Carbon dioxide
 - Light
 - Oxygen

Where are ATP & NADPH Used?

17. Which one of the following products diffuses out of chloroplast during photosynthesis? (2024 Re)
- ADP
 - NADPH
 - O₂
 - ATP
18. Observe the given figure. Identify the different stages labelled with alphabets by selecting the **correct** option. (2024 Re)



- A-Carboxylation, B-Regeneration, C-Reduction
 - A-Reduction, B-Decarboxylation, C-Regeneration
 - A-Carboxylation, B-Reduction, C-Regeneration
 - A-Reduction, B-Carboxylation, C-Regeneration
19. Which of the following are required for the dark reaction of photosynthesis? (2024)
- Light
 - Chlorophyll
 - CO₂
 - ATP
 - NADPH
- Choose the correct answer from the options given below:
- C, D and E only
 - D and E only
 - A, B and C only
 - B, C and D only

20. How many molecules of ATP and NADPH are required for every molecule of CO₂ fixed in the Calvin cycle? (2024)
- 3 molecules of ATP and 3 molecules of NADPH
 - 3 molecules of ATP and 2 molecules of NADPH
 - 2 molecules of ATP and 3 molecules of NADPH
 - 2 molecules of ATP and 2 molecules of NADPH
21. How many ATP and NADPH, are required for the synthesis of one molecule of Glucose during Calvin cycle? (2023)
- 18 ATP and 16 NADPH₂
 - 12 ATP and 12 NADPH₂
 - 18 ATP and 12 NADPH₂
 - 12 ATP and 16 NADPH₂
22. When one CO₂ molecule is fixed as one molecule of triose phosphate, which of the following photochemically made, high energy chemical intermediates are used in the reduction phase? (2022 Re)
- 2 ATP + 2 NADPH
 - 1 ATP + 1 NADPH
 - 1 ATP + 2 NADPH
 - 2 ATP + 1 NADPH

23. Which of the following is **not** a product of light reaction of photosynthesis? (2018)
- ATP
 - NADH
 - NADPH
 - Oxygen

C₄ Pathway & Photorespiration

24. Given below are two statements: (2024)
- Statement-I:** In C₃ plants, some O₂ binds RuBisCO, hence CO₂ fixation is decreased.
- Statement-II:** In C₄ plants, mesophyll cells shows very little photorespiration while bundle sheath cells do not show photorespiration.
- In the light of the above statements, choose the correct answer from the options given below:
- Statement-I is true but Statement-II is false
 - Statement-I is false but Statement-II is true
 - Both Statement-I and Statement-II are true
 - Both Statement-I and Statement-II are false
25. Given below are two statements: (2023-Manipur)
- Statement-I:** RuBisCO is the most abundant enzyme in the world.
- Statement-II:** Photorespiration does not occur in C₄ plants.
- In the light of the above statements, choose the **most appropriate** answer from the options given below:
- Statement-I** is correct but **Statement-II** is incorrect
 - Statement-I** is incorrect but **Statement-II** is correct
 - Both **Statement-I** and **Statement-II** are correct
 - Both **Statement-I** and **Statement-II** are incorrect
26. What is the role of large bundle sheath cells found around the vascular bundles in C₄ plants? (2022)
- To protect the vascular tissue from high light intensity
 - To provide the site for photorespiratory pathway
 - To increase the number of chloroplast for the operation of Calvin cycle
 - To enable the plant to tolerate high temperature

27. Given below are two statements: (2022)
Statement-I: The primary CO_2 acceptor in C_4 plants is phosphoenolpyruvate and is found in the mesophyll cells.
Statement-II: Mesophyll cells of C_4 plants lack RuBisCO enzyme
 In the light of the above statements, choose the correct answer from the options given below.
- Statement-I is incorrect but Statement-II is correct
 - Both Statement-I and Statement-II are correct
 - Both statement-I and statement-II are incorrect
 - Statement-I is correct but Statement-II is incorrect
28. The first stable product of CO_2 fixation in *Sorghum* is: (2021)
- Oxaloacetic acid
 - Succinic acid
 - Phosphoglyceric acid
 - Pyruvic acid
29. The oxygenation activity of RuBisCO enzyme in photorespiration leads to the formation of: (2020)
- 1 molecule of 3-C compound
 - 1 molecule of 6-C compound
 - 1 molecule of 4-C compound and 1 molecule of 2-C compound
 - 2 molecules of 3-C compound
30. Which of the following statements is incorrect? (2020-Covid)
- In C_4 plants, the site of RuBisCO activity is mesophyll cell
 - The substrate molecule for RuBisCO activity is a 5-carbon compound
 - RuBisCO action requires ATP and NADPH
 - RuBisCO is a bifunctional enzyme

31. Phosphoenol pyruvate (PEP) is the primary CO_2 acceptor in: (2017-Delhi)
- C_3 plants
 - C_4 plants
 - C_2 plants
 - C_3 and C_4 plants
32. The process which makes major difference between C_3 and C_4 plants is: (2016 - II)
- Photorespiration
 - Respiration
 - Glycolysis
 - Calvin cycle
33. A plant in your garden avoids photorespiratory losses, has improved water use efficiency, shows high rates of photosynthesis at high temperatures and has improved efficiency of nitrogen utilisation. In which of the following physiological groups would you assign this plant? (2016 - I)
- C_3
 - C_4
 - CAM
 - Nitrogen fixer

Factors Affecting Photosynthesis

34. With reference to factors affecting the rate of photosynthesis, which of the following statements is **not** correct? (2017-Delhi)
- Light saturation for CO_2 fixation occurs at 10% of full sunlight
 - Increasing atmospheric CO_2 concentration upto 0.05% can enhance CO_2 fixation rate
 - C_3 plants responds to higher temperatures with enhanced photosynthesis while C_4 plants have much lower temperature optimum
 - Tomato is a greenhouse crop which can be grown in CO_2 - enriched atmosphere for higher yield

Answer Key

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|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (a) | 3. (c) | 4. (b) | 5. (b) | 6. (d) | 7. (b) | 8. (c) | 9. (b) | 10. (b) |
| 11. (b) | 12. (b) | 13. (c) | 14. (c) | 15. (d) | 16. (a) | 17. (c) | 18. (c) | 19. (a) | 20. (b) |
| 21. (c) | 22. (a) | 23. (b) | 24. (a) | 25. (c) | 26. (c) | 27. (b) | 28. (a) | 29. (a) | 30. (a) |
| 31. (b) | 32. (a) | 33. (b) | 34. (c) | | | | | | |